

MAUVE-MUSE: When Metallicity Follows or Fights Star Formation

A Mass Dependent Inversion in Virgo Spirals

Rongjun Huang (黄镭钧) et al. · MAUVE Collaboration · ICRAR / UWA



INTRODUCTION

Resolved Mass-Metallicity Relation (rMZR) probes the balance of **metal enrichment vs dilution** in discs.

Key question: Does gas-phase metallicity (Z_{gas}) **correlate or anti-correlate** with Star formation rate (SFR) surface density (Σ_{SFR}) at fixed stellar mass surface density (Σ_*), and does the sign changes across discs?

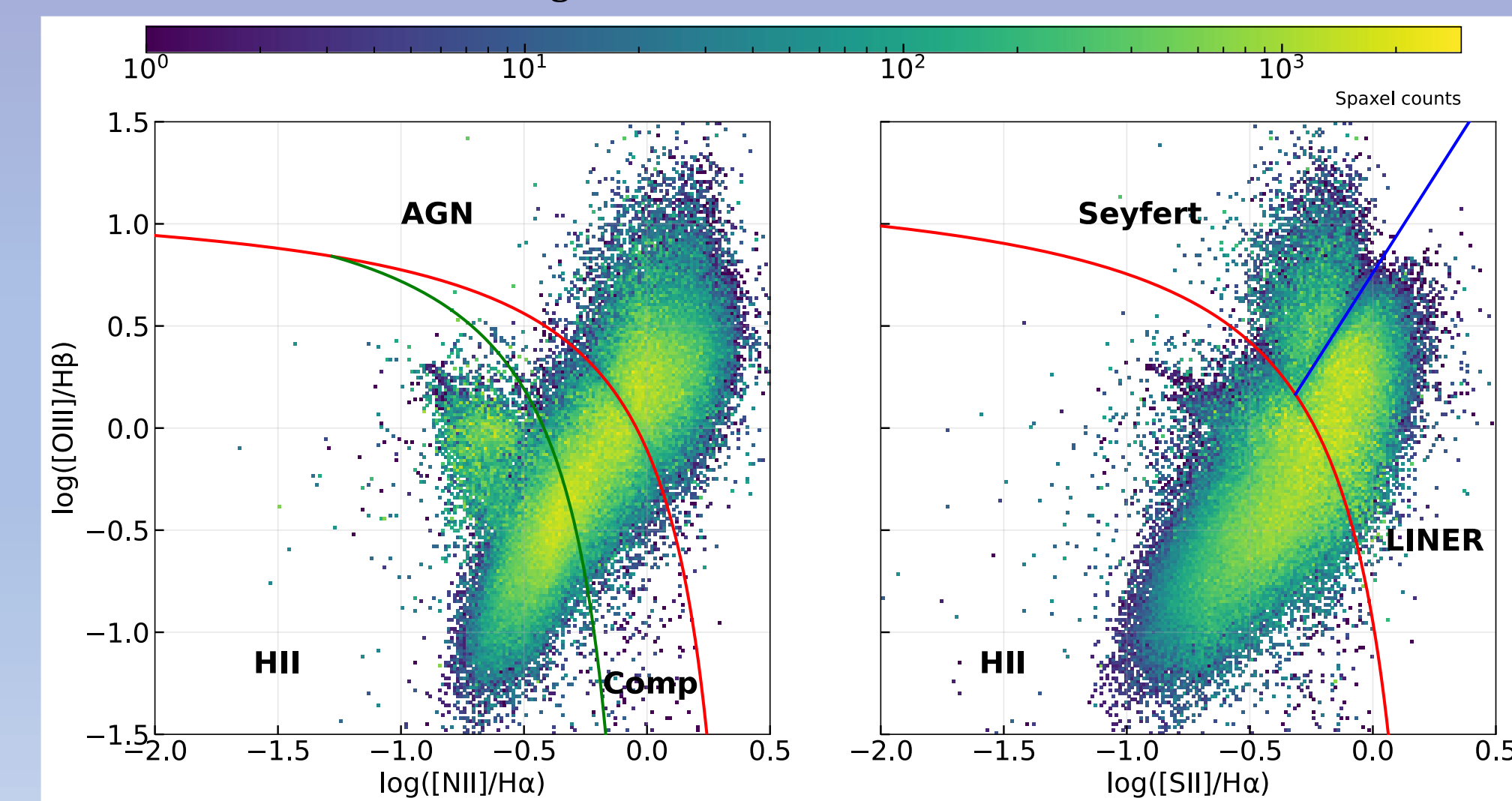
DATA & METHODS

Data: MAUVE-MUSE (Virgo, 100 pc)

- Instrument:** VLT/MUSE wide-field IFU (optical IFS), delivering stellar mass and emission-line maps on ~ 100 pc scales in Virgo spirals.
- Sample:** **14 Virgo spirals** (MAUVE-MUSE v2 internal release); survey observations are **ongoing**.
- Working set:** fiducial ensemble uses **13 galaxies** (excluding the strong outlier **NGC 4383**).
- Resolved Maps used (co-spatial spaxels):** Σ_* , Σ_{SFR} and Z_{gas} .

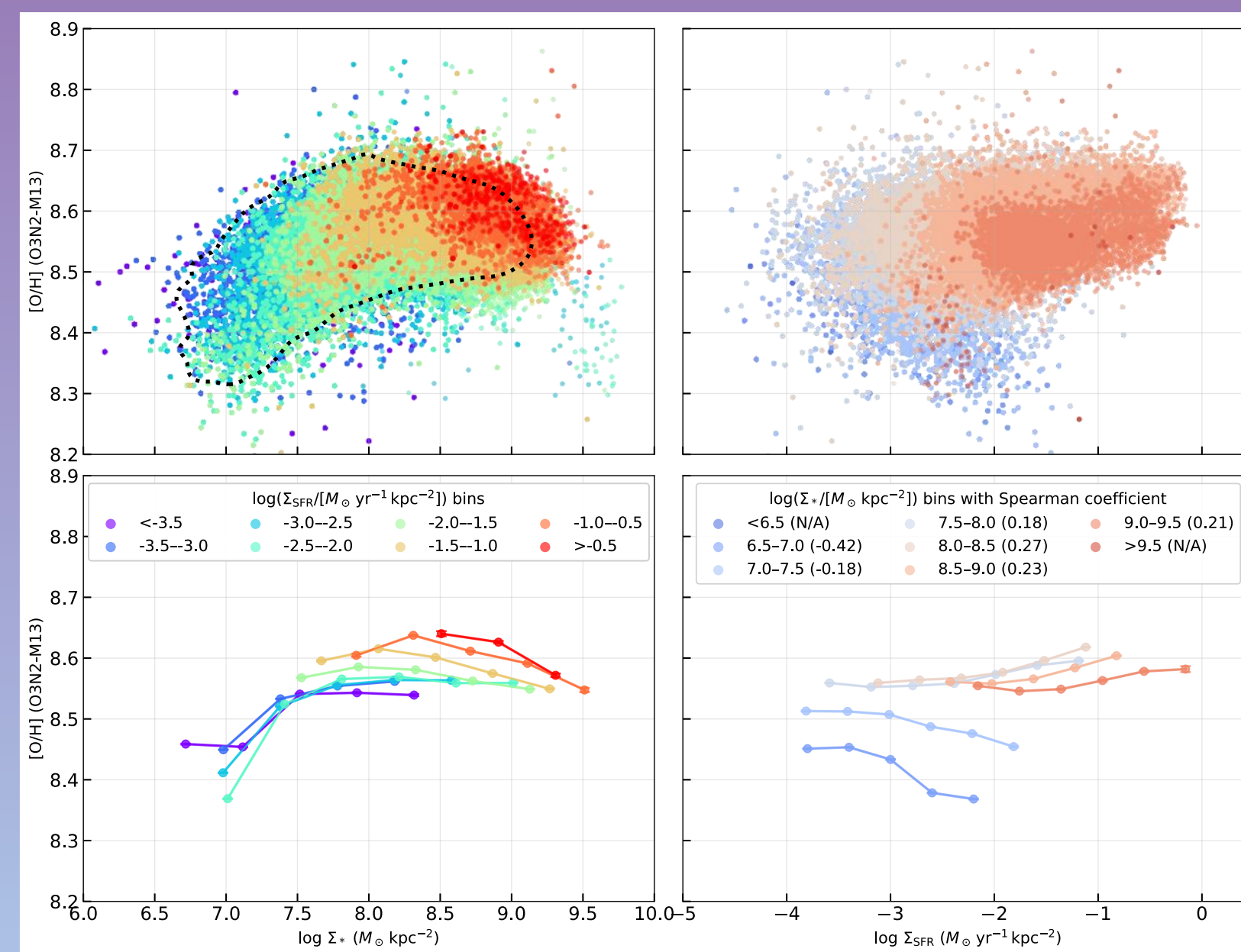
Method: isolate star-forming spaxel and measure Z_{gas}

- Star-Forming selection:** keep H II spaxels using [N II] and [S II] BPT selection diagnostic cuts
- Calibration:** fiducial Z_{gas} from **O3N2-M13** (Marino+2013) using strong lines (dust-corrected H α H β , [O III], [N II])
- $\Delta Z_{gas} - \Delta \Sigma_{SFR}$ coupling measurement: remove primary mass-dependent trends, then measure residual coupling $\rho(\Sigma_*) = \text{corr}(\Delta Z_{gas}, \Delta \log \Sigma_{SFR})|_{\Sigma_*}$

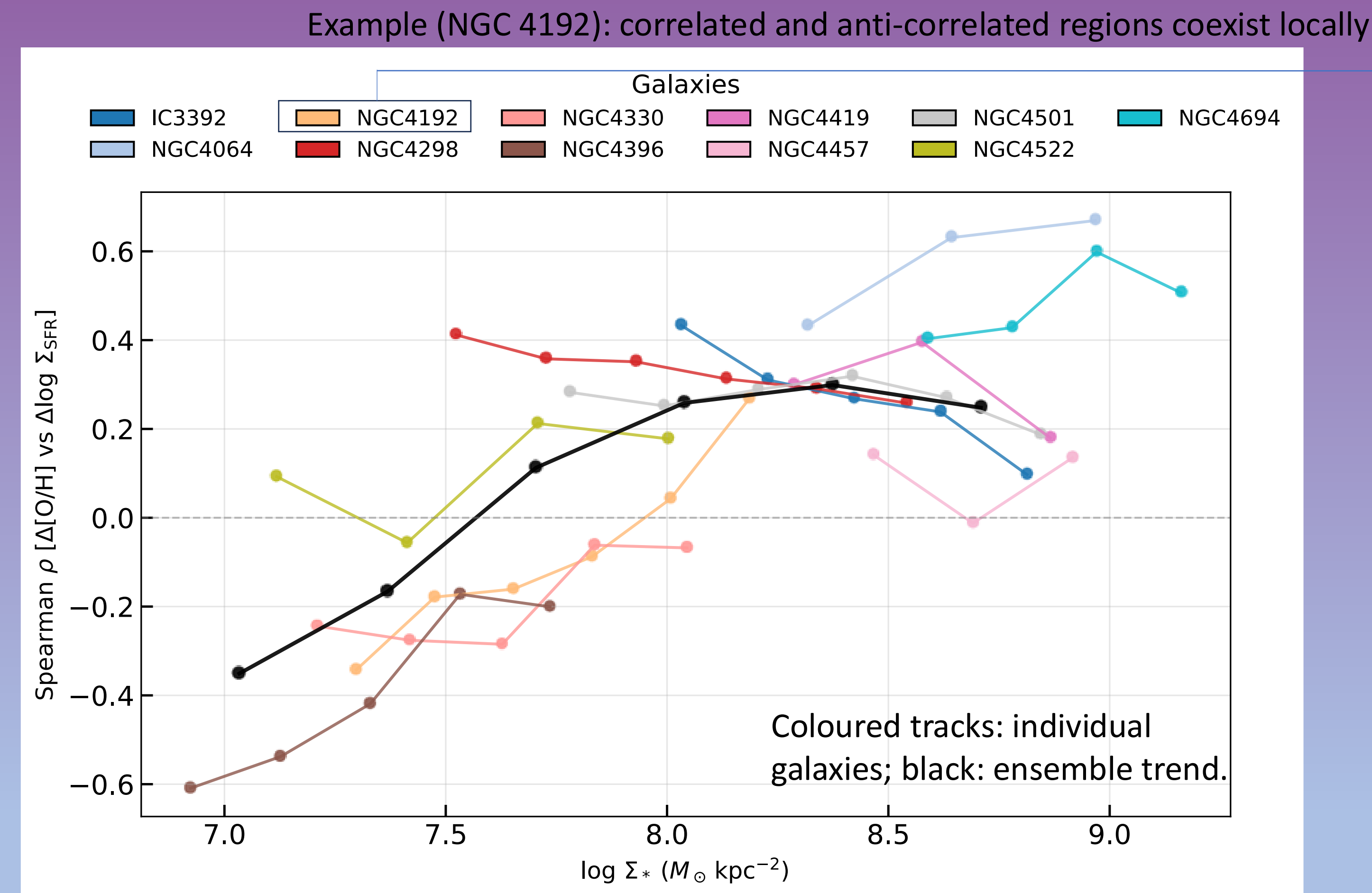


RESULTS

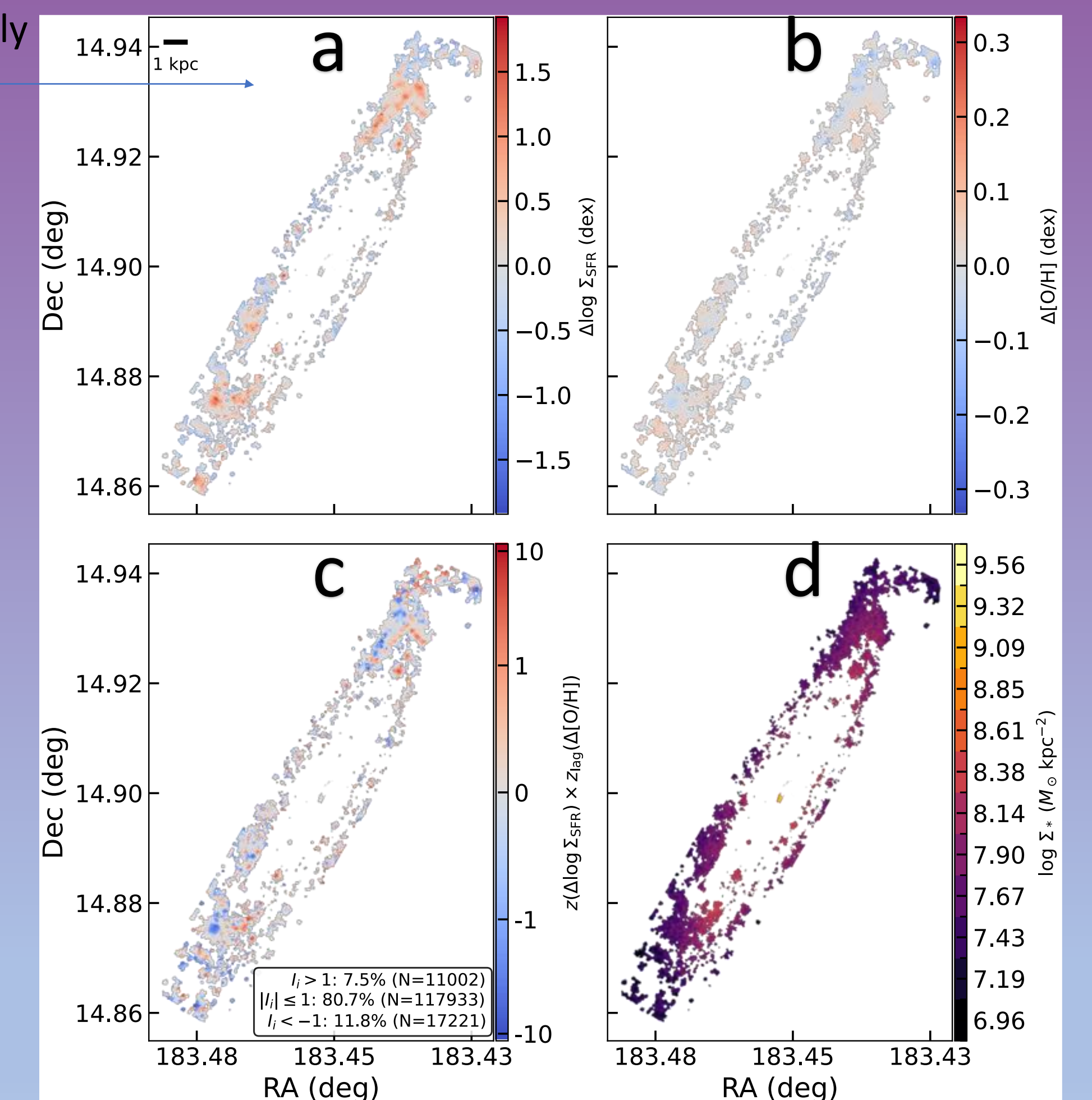
rMZR is well-defined: the *secondary* $\Delta Z_{gas} - \Delta \Sigma_{SFR}$ relation depends on Σ_*



Left: rMZR: Z_{gas} rises with Σ_* ; colour shows Σ_{SFR}
Right: within narrow Σ_* bins, the $Z_{gas} - \Sigma_{SFR}$ trend changes sign: **low Σ_* anti-correlation** → **high Σ_* positive correlation**.



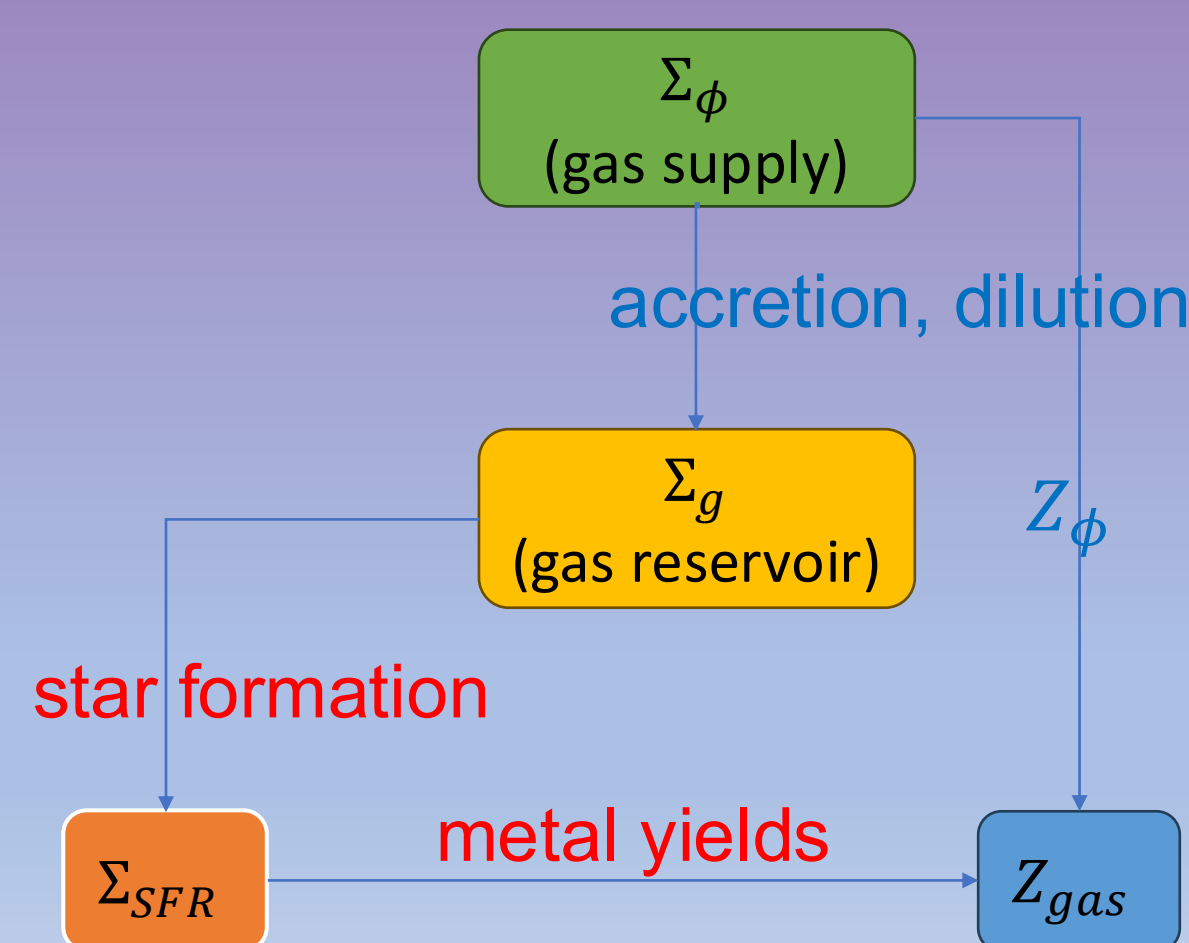
We compute a Spearman rank correlation between residual $\Delta Z_{gas} - \Delta \Sigma_{SFR}$. Across galaxies, Σ_* transitions smoothly from **negative** (gas-supply-dominated) to **positive** (star-formation-dominated), with an inversion point at $\log(\Sigma_*/M_\odot \text{kpc}^{-2}) \approx 7.5 - 8.0$



Panels a,b,d: Σ_{SFR} , Z_{gas} and Σ_* maps. Panel c: A local Moran-like index quantifies neighbourhood co-variation between ΔZ_{gas} and $\Delta \Sigma_{SFR}$. **red** = concordant patches, **blue** = discordant patches.

INTERPRETATION: LOCAL GAS-REGULATOR

$$\frac{d}{dt} \Sigma_g = \Sigma_\phi - (1 - R) \Sigma_{SFR} - \Sigma_{out}; \quad \frac{d}{dt} Z_{gas} = \frac{1}{\Sigma_g} [y(1 - R) \Sigma_{SFR} + (Z_\phi - Z_{gas}) \Sigma_\phi]$$



A spatially resolved *bathtub* model to describe the local gas regulation.
 At fixed Σ_* :

$\Delta Z_{gas} \propto \Delta \log \Sigma_{SFR} - \Delta \log \Sigma_\phi$
Star-formation-driven variability dominates $\Rightarrow \uparrow \Sigma_{SFR}$ (metal yields) with smaller changes in $\Sigma_\phi \Rightarrow \uparrow Z_{gas}$.
Gas-supply-driven variability dominates $\Rightarrow \uparrow \Sigma_{SFR}$ with larger changes in Σ_ϕ (Z_ϕ dilution) $\Rightarrow \downarrow Z_{gas}$.

CONCLUSIONS

- Detect** a mass-dependent inversion in residual $\Delta Z_{gas} - \Delta \Sigma_{SFR}$ coupling
- Show** that this flip-sign is spatially varying in the local Moran-like map
- Interpret** as a transition from **gas supply/dilution-dominated** to **star formation/enrichment-dominated** regulation
- Implication:** this resolved trends unify with integrated metallicity-SFR behaviour.

ROBUSTNESS

- The **sign flip persists** across O3N2-based calibrations;
- HII+Comp BPT selection **keeps the same Σ_* transition**;
- Metallicity outlier, NGC4383, points out a **potential breakdown** in outflows/extraplanar components and/or strong, recent gas-supply perturbations.

CONTACT INFORMATION

Email: Rongjun.Huang@icrar.org
 Personal Website: www.rongjun-huang.com
 MAUVE Website: mauve.icrar.org